Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy under stated assumptions.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pendulum” and “potential energy” to compare your model or result. Sentence starters are optional.

pendulum: an object that swings about a fixed support

potential energy: energy associated with the configuration of interacting objects

kinetic energy: energy associated with motion

01 Warm-up: bring back an earlier idea

Gravitational potential energy changes from -15 J to -3 J. **Find** ΔU and state whether energy is gained or lost.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $mg\,\Delta h = \frac{1}{2}mv^2$; $v = \sqrt{(2\,g\,\Delta h)}$.
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest 0.8 m above its lowest point reaches speed $\sqrt{(2\,gh)}=4$ m/s there with g=10 and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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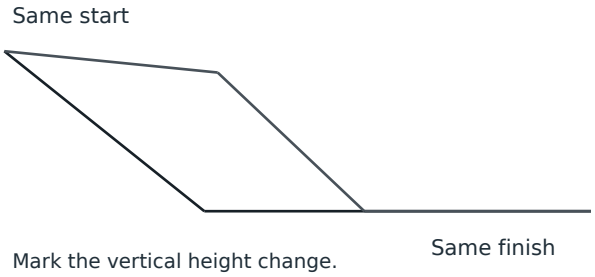
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Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two frictionless tracks start and end at the same heights but take different paths.
Compare final speeds from rest.

Sentence starter: Both … . They differ in … because … .



Mark the vertical height change.

Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart drops 3 m.
Use g=10.
Find lost potential energy and gained kinetic energy.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A classmate uses total ramp length as h in mgh.
Explain how to identify the correct measurement.

Sentence starter: My claim is … . The diagram/data show …, so … .

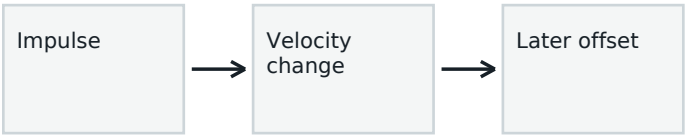
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A fixed impulse changes a target’s sideways velocity by 0.04 m/s.
Find the modeled shift after 600 s and state a limitation.
Label the quantities in the model and state its assumptions.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A pendulum loses 12 J of gravitational energy while moving down. Neglect losses. What is its kinetic energy gain?
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 1 kg pendulum bob is released from rest 0.8 m above the bottom, but reaches only 0.6 m on the opposite side.
Use g = 10 N/kg. Complete the turning-point energy account and find the decrease in mechanical energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3